9.2 Infinite Series

An infinite series is the sum of an infinite sequence of numbers

$$a_1 + a_2 + \cdots + a_n + \cdots$$

For example, given the sequence $\left\{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \cdots\right\} = \{a_n\}_{n=1}^{\infty}$

with $a_n = (1/2)^{n-1}$, we can generate the infinite series

$$\sum_{n=1}^{\infty} a_n = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots .$$

Definition: Given a sequence $\{a_n\}$, the **infinite series** (or simply the **series**) is an expression of the form

$$a_1 + a_2 + \ldots + a_n + \ldots$$

The number a_n is the n**th term** of the series. The sequence $\{s_n\}$ defined by

$$s_1 = a_1$$
$$s_2 = a_1 + a_2$$
$$\vdots$$
$$s_n = a_1 + a_2 + \ldots + a_n = \sum_{k=1}^{n} a_k$$
$$\vdots$$

is the **sequence of partial sum** of the series, the number s_n being the n**th partial sum**. If the sequence of partial sums converges to a limit L, we say that the series **converges** and that its **sum** is L. In this case, we also write

$$a_1 + a_2 + \ldots + a_n + \ldots = \sum_{n=1}^{\infty} a_n = L.$$

If the sequence of partial sums of the series does not converge, we say that the series **diverges**.

Convenient Symbols:

$$\sum_{n=1}^{\infty} a_n \quad \text{or} \quad \sum_{k=1}^{\infty} a_k \quad \text{or simply} \quad \sum a_n$$

Warning: <u>"sequence" and "series"</u>
A series is itself a sequence $\{s_n\}$ obtained from a given sequence $\{a_n\}$ according to the above definition.

Geometric Series

Example: Consider the sequence $\{r^{n-1}\}_{n=1}^{\infty}$ where r is a fixed real numbers, which generates the **geometric series**: $\sum_{n=1}^{\infty} r^{n} = 1 + r + r^{2} + \ldots + r^{n-1} + \ldots$

If $|r| < 1$, the geometric series $a + ar + \ldots + ar^{n-1} + \ldots$ converges to $a/(1 - r)$:

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}, \ |r| < 1.$$

If $|r| \geq 1$, the series diverges.

Example:

1. $a = 1/9$ and $r = 1/3$

2. $\sum_{n=0}^{\infty} \frac{(-1)^n 5}{4^n}$

Further Examples

Example: Consider the series **generated** by the sequence $\{(-1)^n\}_{n=1}^{\infty}$; that is, the series:

Example: The use of "telescoping"

Consider the series

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \frac{1}{3\cdot 4} + \ldots$$

The nth Term Test for a Divergent Series

Example: The series

$$\sum_{n=1}^{\infty} \frac{n+1}{n} =$$

Theorem: If $\sum a_n$ converges, then $a_n \to 0$.

The nth Term Test for Divergence

$\sum_{n=1}^{\infty} a_n$ diverges if $\lim_{n\to\infty} a_n$ fails to exist or is different from zero.

Examples of divergent series:

1. $\sum_{n=1}^{\infty} n^2$

2. $\sum_{n=1}^{\infty} \frac{n+1}{n}$

3. $\sum_{n=1}^{\infty} (-1)^{n+1}$

4. $\sum_{n=1}^{\infty} \frac{-n}{2n+5}$

Another divergent series:

The series

$$1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \ldots + \frac{1}{2^k} + \frac{1}{2^k} + \ldots + \frac{1}{2^k} + \ldots$$

Corollary*: Let $\{a_n\}$ be a sequence of nonnegative real numbers. Then the series $\sum a_n$ converges if and only if the sequence $\{s_k\}$ of partial sums is bounded.

Combining Series

Theorem: If $\sum a_n = A$ and $\sum b_n = B$ are convergent series, then

1. Sum Rule: $\sum(a_n + b_n) = \sum a_n + \sum b_n = A + B$

2. Difference Rule: $\sum(a_n - b_n) = \sum a_n - \sum b_n = A - B$

3. Constant Multiple Rule: $\sum ka_n = k\sum a_n = kA$ for any k.

Example:

1. $\sum_{n=1}^{\infty} \frac{3^{n-1}-1}{6^{n-1}} =$

2. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{3}{2^n} =$

Corollary:

1. Every nonzero constant multiple of a divergent series diverges.

2. If $\sum a_n$ converges and $\sum b_n$ diverges, then $\sum(a_n + b_n)$ and $\sum(a_n - b_n)$ both diverge.

Remarks:

1. Adding or deleting finite terms to or from a series will not affect the series' convergence or divergence. If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=k}^{\infty} a_n$ converges for any natural number k. Conversely, if $\sum_{n=k}^{\infty} a_n$ converges for any $k \geq 1$, then $\sum_{n=1}^{\infty} a_n$ converges.

2. Reindexing any series in such a way that the order of its terms is preserved will not alter its convergence. For example, we can write the geometric series $\sum_{n=1}^{\infty} \frac{1}{2^{n-1}} = 1 + \frac{1}{2} + \frac{1}{4} + \ldots$ as

$$\sum_{n=0}^{\infty} \frac{1}{2^n}, \sum_{n=5}^{\infty} \frac{1}{2^{n-5}} \text{ or even } \sum_{n=-4}^{\infty} \frac{1}{2^{n+4}}.$$

The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

Proof by contradiction:

The 2-series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent.

(Apply Corollary*)

Extra Examples:

1. Find the nth partial sum of the series

$$1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \ldots + (-1)\frac{1}{2^{n-1}} + \ldots$$

2. $\sum_{n=0}^{\infty} \frac{5}{2^n} + \frac{1}{3^n} =$

3. $\sum_{n=1}^{\infty} \frac{40n}{(2n-1)^2(2n+1)^2} =$

4. $\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)} =$